

AI-Powered Customer Intake & Triage Workflow

Architecture Report

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1. Executive Summary

This project implements an AI-powered customer intake and triage workflow for ArcVault using n8n, Google Gemini, and Google Sheets. The solution automates ticket classification, information extraction, escalation handling, queue routing, and structured storage of results.

2. Solution Architecture

Workflow: Customer Ticket → Gemini AI Analysis → Structured JSON Output → Escalation Logic → Queue Assignment → Google Sheets Storage.

3. AI Classification & Enrichment

Google Gemini classifies incoming requests into Bug Report, Feature Request, Billing Issue, Technical Question, and Incident/Outage categories. It also extracts priority, confidence score, identifiers, urgency signals, and a concise summary.

4. Escalation Logic

Tickets are escalated when confidence is below 70%, when incidents/outages are detected, when billing-related issues require review, or when indicators suggest broad customer impact.

5. Queue Routing Strategy

Bug Reports → Engineering
Feature Requests → Product
Billing Issues → Billing
Technical Questions → IT/Security
Incident/Outage → Engineering

6. Data Storage

All processed tickets are stored in Google Sheets including source, category, priority, confidence score, identifiers, routing queue, escalation status, and summary.

7. Technology Stack

n8n, Google Gemini 2.5 Flash, Google Sheets, GitHub, and Loom.

8. Future Enhancements

Future improvements include CRM integrations, sentiment analysis, advanced escalation scoring, analytics dashboards, and human feedback loops.

9. Conclusion

The solution demonstrates how AI can automate customer intake and triage while preserving human oversight for critical scenarios.